Notes:

In my solution to this problem, I actually use an advanced theorem from number theory. The theorem says that

$$\pi(n) < \frac{1.25506n}{\ln(n)},$$

where $\pi(n)$ is the prime-counting function, which denotes the number of primes less than or equal to n. The theorem is supported by proof; it was proven in 1962 by Rosser and Schoenfeld.

You can find their proof at the following websites:

1. https://projecteuclid.org/journals/illinois-journal-of-mathematics/volume-6/issue-1/Approximate-formulas-for-some-functions-of-prime-numbers/10.1215/ijm/1255631807.full
2. http://denise.vella.chemla.free.fr/Rosser-Schoenfeld-1962.pdf
3. https://scispace.com/pdf/approximate-formulas-for-some-functions-of-prime-numbers-c2u470dp6o.pdf

I thought it would help to cite them; otherwise, it looks like I'm waving a magic wand. Number theory is a rich and fascinating subject. The prime-counting function $\pi(n)$ is one of my favorite functions from number theory.

The theorem I refer to is actually corollary 1 in section 3 of the publication, on page 69.